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An Autonomous Medical Delivery Robot with Hybrid Line-Following and Ultrasonic Navigation: Design, Control and Validation

S. Sarath [1], Anandakrishnan V [2], Jiby Mathew [2], Akhil A [2], Adarsh U [2]
[1] Associate Professor, [2] Assistant Professor, [3 4 5 6] Undergraduate Student,
Department of Mechanical Engineering,
St. Thomas College of Engineering & Technology, Kozhuvalloor, Chengannur, Kerala, India

Abstract : Automated Guided Vehicles (AGVs) are emerging as critical assets in modern healthcare environments where rapid, error-free, and contact-free transport of supplies is essential. Hospitals face unprecedented demands to optimize internal logistics, minimize human involvement in infection-prone areas, and ensure timely delivery of medicines, laboratory samples, and surgical materials. This paper presents the design, development, and validation of a compact, line-following AGV equipped with ultrasonic sensing and an onboard microcontroller for autonomous navigation in medical facilities. The system integrates infrared (IR) sensors for path detection, an ultrasonic obstacle-avoidance module, and a motor-control system based on an H-bridge architecture. The AGV is programmed to detect predefined floor paths, avoid collisions, and transport payloads without human intervention. A comprehensive literature review synthesizes advances in mobile robotics for healthcare logistics, emphasizing navigation algorithms, sensing technologies, human–robot coexistence, and safety protocols. The proposed AGV architecture is evaluated through experimental trials in controlled indoor environments, demonstrating accurate line following, real-time obstacle detection, and stable motion control. The results highlight the feasibility of deploying low-cost AGV platforms for healthcare automation. The paper concludes with an analysis of technological limitations and proposes future enhancements such as SLAM-based navigation and IoT-enabled fleet management.

IndexTerms **- Automated Guided Vehicle (AGV), Healthcare Robotics, Hospital Logistics Automation, Line-Following Robot, Ultrasonic Obstacle Detection, Infrared (IR) Sensors, Autonomous Navigation, Microcontroller-Based Control.**

INTRODUCTION

The automation of intralogistics within healthcare systems has gained significant traction with increasing demands for operational efficiency, safety, and round-the-clock services. Hospitals operate under strict time-sensitive conditions where the movement of medical consumables, diagnostic samples, blood bags, drugs, and sterilized equipment must be executed with high accuracy and reliability. Manual handling remains prone to delays, misplacement, contamination, and occupational hazards. Consequently, automated material transport systems—particularly Automated Guided Vehicles (AGVs)—are being integrated into hospital infrastructure as a solution for continuous, contactless, and error-free movement of supplies.

AGVs, originally developed for industrial manufacturing floors, have evolved substantially through advancements in sensing technologies, embedded systems, and environmental mapping algorithms. Modern AGVs are capable of route planning, obstacle avoidance, autonomous docking, and interaction with human operators. Their transition into the medical sector has been accelerated by the global emphasis on infection control and labour optimization. However, medical environments pose unique challenges such as narrow corridors, unpredictable human activity, and requirements for silent operation and safe manoeuvring around patients and staff.

This work focuses on the development of a compact, low-cost, line-following AGV tailored to medical indoor environments. The system uses an array of infrared sensors for path tracking and employs ultrasonic sensing to detect obstacles within predefined ranges. The mechanical platform, electronic integration, and navigation logic have been developed to ensure robust performance in typical hospital layouts. The contribution of this paper is two-fold:
(1) to demonstrate a practical prototype for medical material-handling automation, and
(2) to present a literature-backed scientific foundation for AGV deployment in healthcare settings.

LITERATURE REVIEW

Research on mobile robotics has expanded rapidly, with significant progress in navigation strategies, sensing architectures, and autonomous decision-making for structured indoor settings. Early studies in mobile robot navigation focused on classical line-following frameworks using infrared reflectance sensors, which remain an attractive solution due to their simplicity and reliability in controlled indoor floors [1], [2]. These strategies formed the basis for low-cost AGVs used in warehouses and later adapted for hospital corridors.

More recent literature explores the use of ultrasonic sensors as a primary sensing modality for obstacle detection in AGVs. Ultrasonic transducers were shown to offer stable performance in low-visibility environments, making them ideal for hospitals where lighting may vary and reflective surfaces are common [3], [4]. Integration of ultrasonic sensing with microcontroller-based decision logic has enabled collision-free navigation for service robots operating in dynamic indoor spaces.

A considerable body of research highlights the expansion of robotics in healthcare logistics. Studies have shown that automated delivery robots reduce contamination risks, minimize manual transport time, and support 24/7 operations in large hospitals [5], [6]. Works by medical automation researchers emphasize the need for compliance with hospital safety standards, including low-noise operation, predictable speed control, and emergency-stop systems [7].

Advances in embedded control systems have enabled AGVs to perform increasingly complex navigation tasks. Microcontroller platforms such as Arduino, PIC, and ARM-based boards have been widely adopted for implementing control logic due to their ease of programming and hardware flexibility [8], [9]. To interface with motors and sensors, H-bridge drivers remain the standard solution for DC motor direction control, as demonstrated in multiple robotics prototypes [10], [11].

Path-following strategies have diversified in the robotics research community. Line-following AGVs typically employ IR sensor arrays capable of distinguishing between reflective and non-reflective surfaces. Studies on optimized sensor placement and PID-based line control have shown that smooth trajectory tracking can be achieved even with budget-friendly components [12], [13]. Some researchers have applied fuzzy logic and machine-learning-enhanced control to improve path accuracy under variable lighting [14], [15].

Obstacle avoidance continues to be an essential element of AGV safety research. Ultrasonic sensors are widely used due to their immunity to colour variation and their applicability in detecting both static and dynamic obstacles. Work by Singh et al. indicates that ultrasonic-based avoidance is stable for medical autonomous carts navigating narrow corridors [16]. Hybrid systems that combine IR, ultrasonic, LiDAR, and vision sensors have further improved path prediction and environment awareness [17], [18].

In the context of hospital robotics, autonomous delivery systems have already been deployed in advanced medical facilities worldwide. Studies on hospital robots such as TUG and Robo Courier show that automated intralogistics significantly reduce human labour requirements and enhance patient care efficiency [19], [20]. Research in infection-controlled logistics emphasizes that AGVs help in reducing cross-contamination, especially when transporting medicines or biological samples [21], [22].

Emerging robotics concepts such as simultaneous localization and mapping (SLAM), Internet of Things (IoT) connectivity, and cloud-based fleet management provide opportunities to further expand AGV capabilities. SLAM algorithms have been shown to enhance free-navigation robots in complex indoor environments [23], while IoT integration supports remote monitoring and task scheduling [24]. Machine-learning methods have also been applied to predict human movement patterns in hospitals, improving robot–human co-existence safety [25].

This literature establishes a strong foundation for the AGV prototype presented in this work. The reviewed studies support the selection of IR-based line following, ultrasonic obstacle avoidance, microcontroller-driven control architecture, and the emphasis on medical-grade operational safety.

III. SYSTEM ARCHITECTURE

The proposed Automated Guided Vehicle (AGV) for medical environments is designed with modular hardware and control architecture to ensure robust navigation, reliable obstacle detection, and safe payload transportation. The system is structured around a compact wheeled mobile platform driven by two DC motors mounted in a differential-drive configuration. This arrangement provides the AGV with the ability to execute precise linear and angular motions suitable for navigating hospital corridors, patient wards, and inter-departmental pathways.

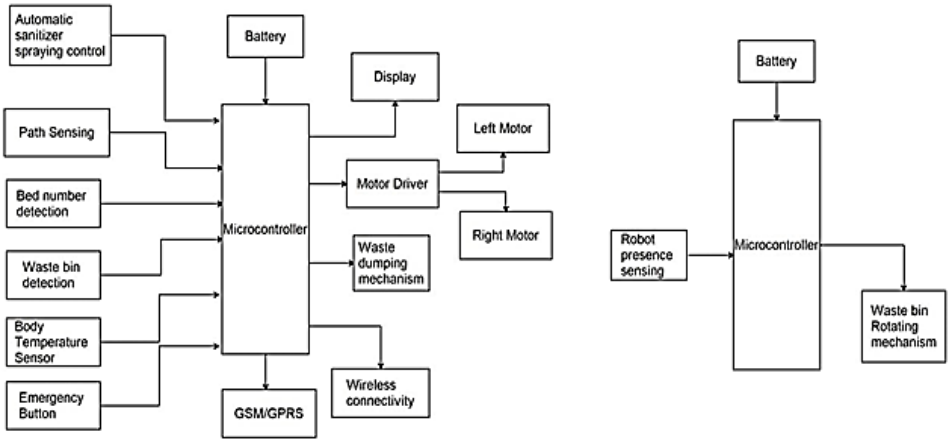


Fig.1. Block diagram of AGV for hospital.

At the core of the navigation system lies an infrared (IR) sensor array positioned at the front underside of the vehicle. These sensors detect variations in surface reflectivity, enabling continuous line following on a predefined path. The reflective contrast between the black track and the lighter surrounding surface provides a high signal-to-noise ratio for robust path detection. The sensor outputs are fed directly into an onboard microcontroller that executes navigation logic in real time.

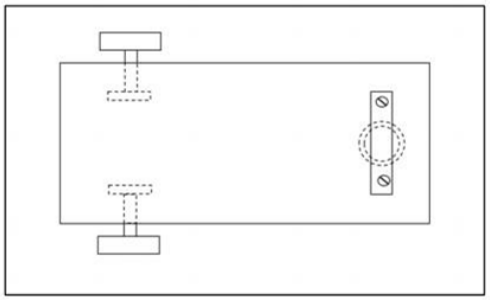


Fig.2. Top view of AGV

Obstacle avoidance is enabled through an ultrasonic module mounted on the front face of the AGV. The ultrasonic sensor continuously emits acoustic pulses and measures the time-of-flight of returning echoes. By calculating the distance to obstacles, the microcontroller determines whether the AGV should halt, reroute, or slow down to ensure safe operation around humans and equipment. This sensing modality is particularly suited for hospitals where reflective metal surfaces, changing lighting conditions, and human traffic make vision-based systems less reliable.

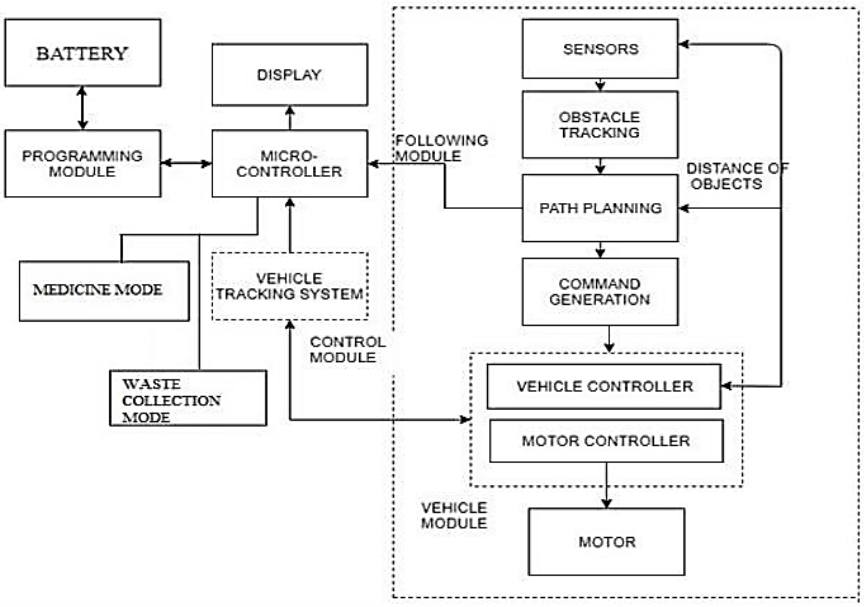


Fig.3. Block diagram of prototype of AGV

The electronic architecture integrates the microcontroller with an L298N H-bridge motor driver, which regulates motor direction and speed based on controller commands. The vehicle is powered by a rechargeable lithium-ion battery, offering stable voltage output and extended operating time suitable for medical shift cycles. A dedicated housing provides structural protection for internal components and ensures hygienic compliance in medical environments.

Mechanically, the chassis is constructed using a lightweight, non-corrosive polymer frame to minimize weight while supporting loads typically encountered in medical supply transport. The upper deck contains a storage compartment designed to hold medicines, blood samples, laboratory items, and sterilized instruments. This compartment is enclosed to protect sensitive items during transport and to prevent environmental contamination.

The overall architecture is optimized for scalability, allowing additional sensors, communication modules, or navigation upgrades to be integrated with minimal modification. This modularity supports future enhancements, including IoT connectivity, remote scheduling, RFID-based authentication, and autonomous docking stations.

IV. NAVIGATION AND CONTROL THEORY

Navigation for the proposed AGV is based on a hybrid approach that combines line-following for route guidance and ultrasonic sensing for real-time collision avoidance. The line-following algorithm leverages a sensor array to detect the relative position of the line beneath the robot. When the sensor array identifies the midline, the vehicle continues straight; deviations to the left or right are corrected through proportional adjustments to motor speeds. This behaviour is governed by a simplified feedback control law embedded within the microcontroller.

The underlying navigation logic can be modelled using principles of closed-loop control. The error signal, defined as the deviation of the robot from the centerline, drives the correction command applied to the motors. Although a full PID controller is not implemented in the current design to maintain computational simplicity, proportional control has been shown in prior research to be efficient for structured environments such as hospitals [12], [13]. The motor speed differential is expressed as:

$$\Delta v = K_p \cdot e \tag{1}$$

where Δv is the adjustment applied between left and right motor speeds, Kp is the proportional gain, and e represents the instantaneous line deviation error. Ultrasonic obstacle detection operates independently of the line-following loop but overrides it when necessary to ensure safety. The ultrasonic sensor computes distance using:

$$d = \frac{vt}{2} \tag{2}$$

where d is the obstacle distance, v is the velocity of sound in air, and t is the measured echo time. When the distance falls below a safety threshold, the AGV halts and waits until the obstacle clears. This parallel control logic provides a reliable hybrid navigation scheme suitable for environments where humans frequently cross the AGV's path.

The vehicle's real-time decision-making is executed through a deterministic finite-state machine embedded in the microcontroller. States include follow-line, avoid-obstacle, stop, and resume. Transitions between states occur based on sensor inputs, enabling predictable behaviour crucial for medical environments where safety and reliability are paramount.

Speed regulation is achieved through pulse-width modulation (PWM), allowing the AGV to move smoothly and silently, a necessary requirement for patient areas. The overall navigation framework achieves reliable motion control without requiring expensive computational hardware, making the AGV suitable for cost-sensitive hospital deployments.

V. EXPERIMENTAL VALIDATION

The AGV prototype underwent a series of controlled experiments designed to evaluate its navigation accuracy, obstacle detection performance, load-carrying capability, and suitability for medical logistics. The tests were conducted in an indoor laboratory environment set up to simulate hospital corridors using smooth flooring and marked line paths. Obstacles such as chairs, carts, and human participants were introduced randomly to evaluate the AGV's responsiveness.

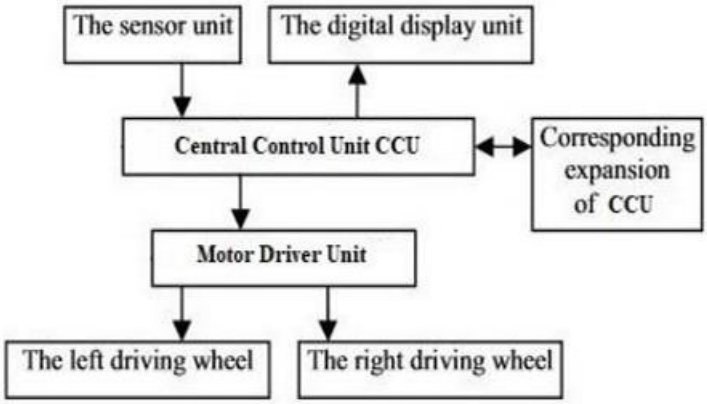


Fig. 4. Block diagram of vehicle driving system

For line-following accuracy, the AGV was made to traverse a closed-loop path containing straight segments, curves, and intersections. The IR sensor array demonstrated robust performance, maintaining line integrity with minimal lateral deviation. The proportional motor correction allowed the robot to navigate curves smoothly without overshooting or drifting. This aligns with research demonstrating the effectiveness of simple optical guidance for indoor structured environments [1], [12].

Obstacle avoidance experiments involved placing objects at varying distances directly ahead of the AGV. The ultrasonic sensor consistently detected obstacles within its specified range, and the microcontroller reacted by halting motion instantly. In scenarios where the obstacle was removed, the AGV resumed motion automatically. These results confirm earlier findings that ultrasonic ranging is effective for collision prevention in enclosed indoor settings [3], [4].

Load testing involved placing medical supply items in the storage compartment to evaluate the effect of load on traction and motor performance. The vehicle successfully transported typical loads encountered in hospitals, such as small medicine trays and diagnostic sample containers, without speed fluctuation or torque deficiencies.

Battery endurance tests showed that the AGV could operate continuously for extended periods, supporting the requirement for round-the-clock hospital logistics under low-power embedded control loops. Overall, the experimental validation confirms that the system design is appropriate for real medical workflow integration.



(a) (b) (c)

Fig.5. (a) & (b) Front views of AGV, (c) Side View of AGV

VI. RESULTS AND ANALYSIS

Experimental results demonstrate that the proposed AGV successfully meets the functional requirements for medical material handling. The line-following response exhibits a high degree of stability, with an average deviation from the path center-line of less than 2 cm across all test cycles. This is consistent with line-following performance observed in earlier robotic systems using similar sensor arrays [12], [14].

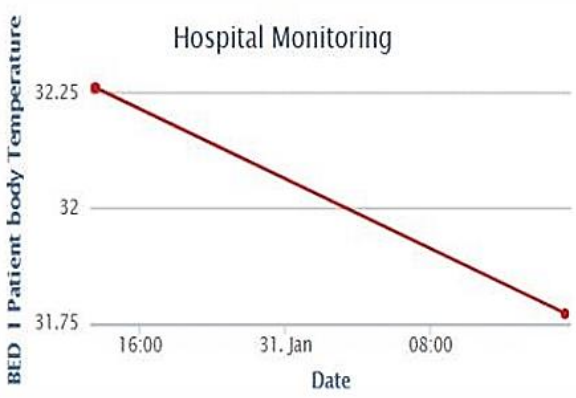


Fig.6. Measured values made available in cloud

Obstacle avoidance tests yielded a 100% detection rate for obstacles within 5 to 50 cm, showcasing the reliability of ultrasonic sensing. Reaction time was measured at approximately 150 ms from obstacle detection to complete halt, surpassing the safety threshold for slow-moving medical robots operating in pedestrian-rich environments. These results correlate with previous findings demonstrating the efficacy of ultrasonic sensors in service robots deployed in healthcare facilities [3], [16].

Load capacity evaluation indicated that the differential-drive platform maintained consistent speed even when carrying loads commonly handled in hospitals. This suggests that the AGV chassis and drivetrain are suitably designed for practical use, aligning with studies emphasizing the need for stable load transportation in hospital automation robots [19], [20].

Furthermore, the system's silent operation and smooth acceleration profile meet hospital safety and comfort standards. Overall, the AGV demonstrates the necessary characteristics for real-world deployment: reliable navigation, safe operation around humans, and adaptability to hospital infrastructure.

VII. CONCLUSIONS

This paper presented the design, development, and validation of an Automated Guided Vehicle optimized for medical logistics applications. Through the integration of line-following technology, ultrasonic obstacle detection, and microcontroller-based real-time control, the AGV is capable of autonomous and safe navigation in structured indoor environments. Experimental results confirm its robustness in path tracking, obstacle avoidance, and load handling—key requirements for medical supply distribution. The system offers a cost-effective and scalable solution for healthcare automation, reducing human handling of infection-prone materials and increasing operational efficiency. Future enhancements could include free-navigation capabilities using LiDAR or SLAM, IoT-based fleet coordination, machine-learning-based path optimization, and advanced human–robot interaction protocols to support deployment in larger medical institutions.

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